Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT1012
Course title (English)	Honours Calculus II
Course title (Chinese)	微积分荣誉课程（二）
Units	3
Language of Instruction	English
Description (English)	This course is a continuation of MAT1011 Calculus (Extended) I. Both courses are designed for those students who are interested in mathematics, and who wish to build a solid foundation for more advanced courses. These courses aimed to develop students' abilities of analyzing, proving and problem-solving. MAT1012 teaches multi-variables calculus with depth. Topics include infinite sequences and series, Taylor expansions, vectors and spaces, partial derivatives and differentiation, double and iterated integrals, triple integrals, integration of vector fields, line and surface integrals, Green's theorem, Stokes' theorem, divergence theorem.
Description (Chinese)	本课程为MAT1011微积分精讲（一）的延续课程。 两门课程都是面向对数学有兴趣的学生，为学习后续课程提供坚实的基础，并培养学生的分析、论证和解决问题的能力。MAT1012讲授多变量微积分的基本原理和方法，内容包括：序列与级数、Taylor展开，向量与空间、偏导数、多元函数的微分、二重积分与累次积分、三重积分、向量场的积分、曲线积分、曲面积分、Green定理、Stokes定理、散度定理。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT1011: Honours Calculus I

B. Co-requisites

None.

3. Learning Outcomes

Upon successful completion of the course, the students will:

• Acquire understanding of infinite sequences and series, power series and Taylor expansions;
• Acquire understanding of the central concepts and the important theorems in multivariable calculus;
• Acquire the ability of performing computations related to differentiation and integration in multivariable calculus;
• Acquire the ability of applying results in multivariable calculus to solve relevant problems in other fields.

4. Course syllabus

Sequences and series: Limits of sequences, some important limits, Cauchy criterion for sequences, bounded monotone sequences, subsequences, least upper bound and greatest lower bound, existence of least upper bound and greatest lower bound of a bounded set, convergent subsequence of a bounded sequence. Convergence of series, properties of convergent series, Cauchy criterion for series, integral test, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, rearrangements and multiplication of absolutely convergent series, Dirichlet test, Abel test, sequence of functions, uniform convergence of sequence of functions. Series of functions, uniform convergence of series of functions, Cauchy criterion for uniform convergence of series of functions, Wererstrass's M-test, properties of uniform convergent series

Power series: Convergence of power series, radial of convergence, properties of power series. Taylor series, Maclaurin series, applications of Taylor series, series solution of ODEs

Parametric equations: Parametrizations of plane curves, calculus with parametric curves. Differentiation in polar coordinates, curves, arc length and area in polar coordinates. Conic sections, conics in polar coordinates

Vectors, vector fields and geometry of spaces: Three-dimensional coordinate systems, vectors, dot products, cross products, lines and planes in spaces, cylinders and quadric surfaces. Differentiation and integration for vector fields of one variable. Curves in space, tangents, arc length, curvature, normal vectors of a curve. Tangential and normal components of acceleration, velocity and acceleration in polar coordinates

Partial differentiation: multivariate functions, limits and continuity in higher dimension, partial derivatives of a multivariable functions, linearization and differentiability in higher dimensions, chain rules, implicit differentiation, directional derivatives and gradients. Extreme values and saddle points, constrained maxima and minima, method of Lagrange multipliers, Taylor's formula in higher dimensions.

Implicit function theorem: implicit functions, implicit function theorem, inverse function theorem, applications: coordinate transforms, tangent lines and normal planes of a curve, tangent planes and normal lines of a surface, constraint extrema

Double integrals: double and iterated integrals over rectangles, Fubini's theorem on a rectangle, rectifiability and area of a planar region, definition of double integrals over a general domain, finite covering lemma, Riemann integrability over a rectifiable region, properties and computation of double integrals, Fubini's theorem on a general region, area and volume by double integration, double integrals in polar form

Integrals depending on parameters: Concept and properties of an integral depending on a parameter, convergence and uniform convergence of an improper integral with a parameter, properties

Triple integrals: Definition of triple integrals, Riemann integrability, evaluation of triple integrals in rectangular coordinates, integrals in cylindrical and spherical coordinates, substitutions in triple integrals, applications

Line integral and vector fields: Line integrals of scalar functions, evaluation and properties.

Surface integrals: Area of surface, surface integrals of scalar functions, evaluation, and applications. Surface integrals of vector fields. Vector fields, definition of line integrals of vector fields, path independence, conservative fields, potential functions, k-component of curl, divergence, Green's theorem and applications

Stokes' theorem and divergence theorem: Area of a surface, surface integrals of scalar functions, evaluation, and applications. Orientation of a surface, surface integrals of vector fields, evaluation and applications, curl of vector fields, Stokes' theorem and applications,
divergence of vector fields in space, divergence theorem and applications, unifying the integral theorems

5. Assessment Scheme

Component/ method	**% weight**
Assignments	20

Quizzes	20
Mid-Term Exam(s)	20
Final Exam	40

6. Grade descriptor

General

Grade	Description
A	Demonstrate an excellent understanding of multivariate differentiation and integration, and demonstrate a strong ability to clearly and logically apply theorems and rules of differentiation and integration to solve problems in calculus. Ability of solving unfamiliar problems is expected.
B	Demonstrate a good understanding of multivariate differentiation and integration, and demonstrate a substantial ability to apply theorems and rules of differentiation and integration to solve problems in calculus.
C	Demonstrate an average understanding of multivariate differentiation and integration, and demonstrate an average ability to apply theorems and rules of differentiation and integration to solve problems in calculus.
D	Demonstrate some understanding of multivariate differentiation and integration, and demonstrate the ability to apply basic methods to solve simple problems in calculus.
F	• Demonstrate a poor understanding of the course materials. • Non-ability in applying basic methods to solve simple problems.

7. Feedback for evaluation

Formal Course and Teaching Evaluation
Informal feedback and/or teaching assistant
School retreat and programme review

8. Reading

Required Reading

1.Thomas' Calculus (13th Edition), ISBN-13 978-1292089799, Thomas Jr., Maurice D. Weir, Joel R. Hass., Pearson, 13TH, United States

9. Course components

Activity	Hours/week
Lecture	5
Tutorial	2

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Limits of sequences, some important limits, Cauchy criterion for sequence, bounded monotone convergence, subsequences, least upper bound and greatest lower bound, existence of least upper bound and greatest upper bound of a bounded set, convergent subsequences of a bounded sequence. Convergence of series, properties of convergent series
2	Cauchy criterion for series, integral test, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, rearrangements and multiplication of absolutely convergent series, Dirichlet test, Abel test. Sequence of functions, uniform convergence of sequence of functions
3	Series of functions, uniform convergence of series of functions, Cauchy criterion of uniform convergence for series of functions, Weierstrass' M-test, properties of uniform convergent series. Convergence of power series, radius of convergence, properties of power series, Taylor series, Maclaurin series, applications, series solution of ODEs
4	Parametric functions, calculus with parametric curves, differentiation in polar coordinates, arc length and area in polar coordinates, conics in polar coordinates. Vectors, dot products, cross products, lines and planes in spaces. Differentiation and integration for vector fields of one variable
5	Curves in space, tangents, arc length, curvature, normal vectors of a curve, tangential and normal components of acceleration, velocity and acceleration in polar coordinates. Sets in plan and in space, multivariate functions, limits and continuity in higher dimension, partial derivatives of a multivariable function
6	Linearization and differentiability in higher dimensions, chain rule, implicit differentiation, directional derivatives, gradients. Extreme values and saddle points, constrained maxima and minima, method of Lagrange multipliers, Taylor's formula in higher dimensions
7	Implicit function theorem, inverse function theorem, applications: coordinate transforms, tangent lines and normal planes of a curve, tangent planes and normal lines of a surface, constraint extrema
8	Double integrals and iterated integrals over rectangles, Fubini's theorem on a rectangle, rectifiability and area of a planar region, definition of double integrals over a general domain, finite covering lemma, Riemann integrability over a rectifiable region
9	Properties and computation of double integrals, Fubini theorem on a general region, area and volume by double integration, substitution in double integrals, double integrals in polar form.

10	Concept and properties of an integral depending on a parameter, convergence and uniform convergence of an improper integral with a parameter, properties
11	Definition of triple integrals, Riemann integrability, evaluation of triple integrals in rectangular coordinates, integrals in cylindrical and spherical coordinates, substitutions in triple integrals, applications. Definition of line integrals of scalar functions, evaluation and properties
12	Vector fields, definition of line integrals of vector fields, path independence, conservative fields, potential functions, k-component of curl, divergence
13	Green's theorem and applications. Area of a surface, surface integrals of scalar functions, evaluation, and applications. Orientation of a surface, surface integrals of vector fields, evaluation and applications, curl of vector fields, Stokes' theorem and applications
14	Divergence of vector fields in space, divergence theorem and applications, unifying the integral theorems

11. Any other information

This course is a substitution of MAT1002 Calculus II. The students who intend to declare for Mathematics and Applied Mathematics are strongly recommended to take this one instead of MAT1002.

12. Version date

Version number	2
As of (date)	2023/12/11